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Library Management System

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1. ABSTRACTION / INTRODUCTION

This project is a desktop-based library management system developed using C#.NET and SQL Server. The application provides functionalities for managing books, registered users, borrowing and returning books, and handling inquiries and reservations. The main goal is to digitize the operations of the library, helping librarians to manage daily tasks efficiently, save time, and avoid manual ledger errors.

2. OBJECTIVE OF THE PROJECT

- To automate the manual, paper-based tasks of the library.
- To provide secure user registration and admin login.
- To allow efficient book loan, return, and reservation handling.
- To support book inquiry and real-time availability checks.
- To automatically calculate late return fine amounts (Rs. 10 per day).

3. TOOLS AND TECHNOLOGIES USED

The following software, frameworks, and languages were used to design, build, and maintain this application:

Component	Tools and Technology Used
Frontend	Windows Forms (C#.NET)
Backend Logic	C#.NET Framework
Database	SQL Server Management Studio
Development IDE	Microsoft Visual Studio

4. FUNCTIONAL REQUIREMENTS

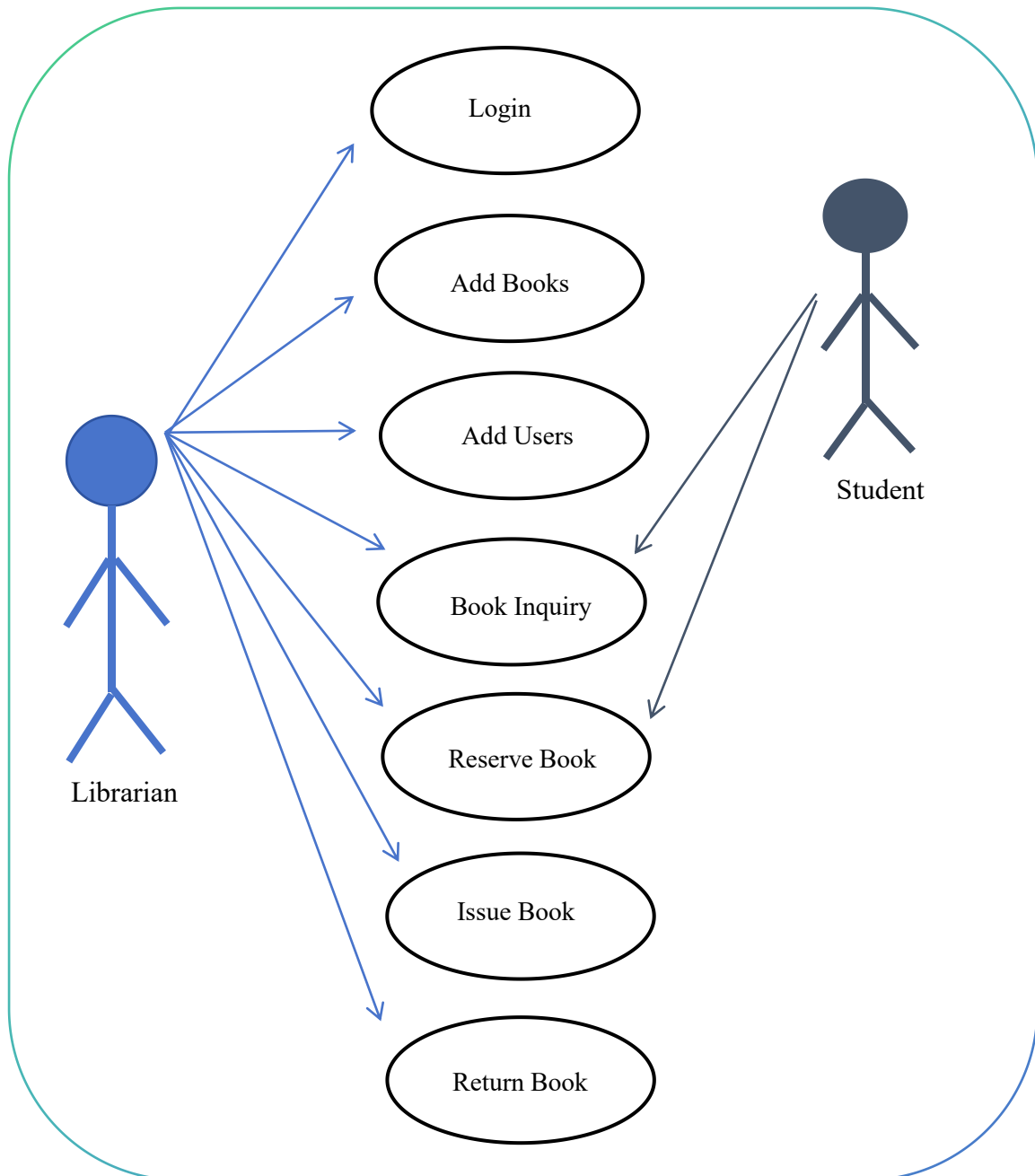
These describe the specific features, functions, and behaviors of the software system:

- Secure Login: Users must log in with a valid admin username and password.
- Member Management: Admin can add new members and view registered users.
- Book Management: Admin can add new books with specific classifications.
- Book Search: The system must allow the librarian to search for books dynamically.
- Loan & Return: System must track issue/due dates and calculate fines for late returns.

5. SYSTEM DESIGN

5.1 Use Case Diagram

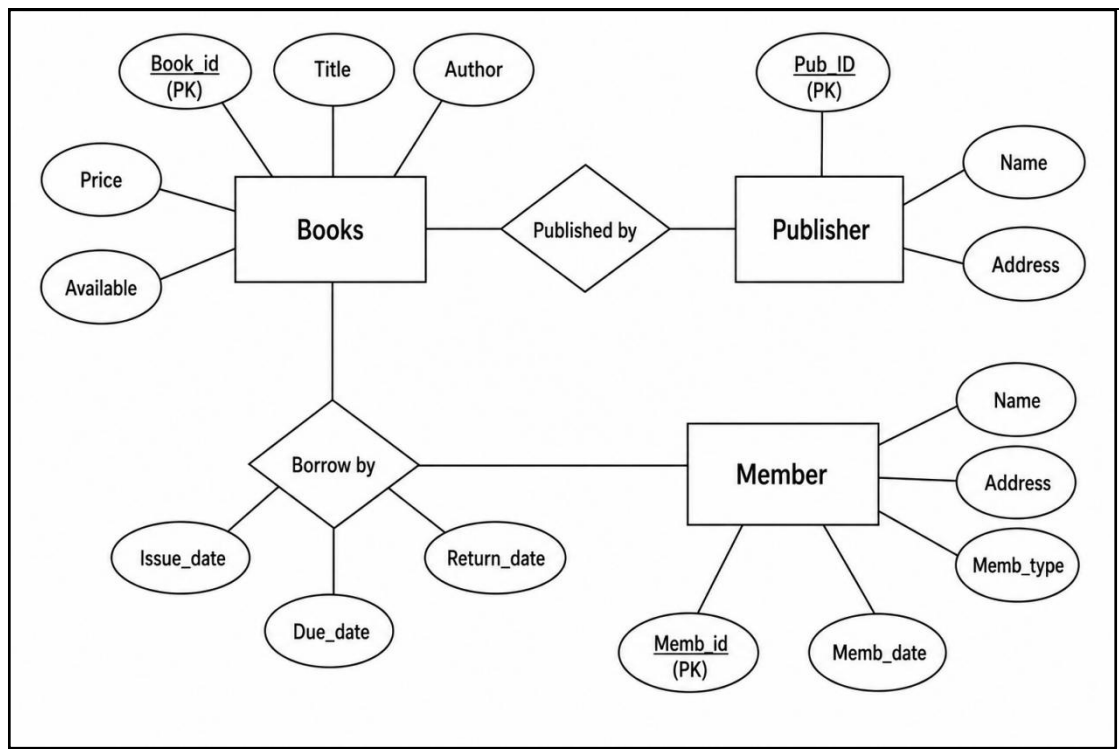
The Use Case diagram shows the interactions between the Librarian (Admin), the Student (Member), and the main system modules.



Case Diagram

5.2 ER Diagram

The Entity-Relationship (ER) Diagram shows the logical database layout and relationships between main tables like Books, Members, and Loans.



ER Diagram

6. PROJECT UI AND EXPLANATION

✧ [Login Page](#)

The screenshot shows a login form with a brown background. It features a user icon with a blue arrow pointing to the right. Below the icon, there are two input fields: "Username : Admin" and "Password : ****". At the bottom, there is a "Login" button. An arrow points from the text "LOGIN BUTTON" to the "Login" button.

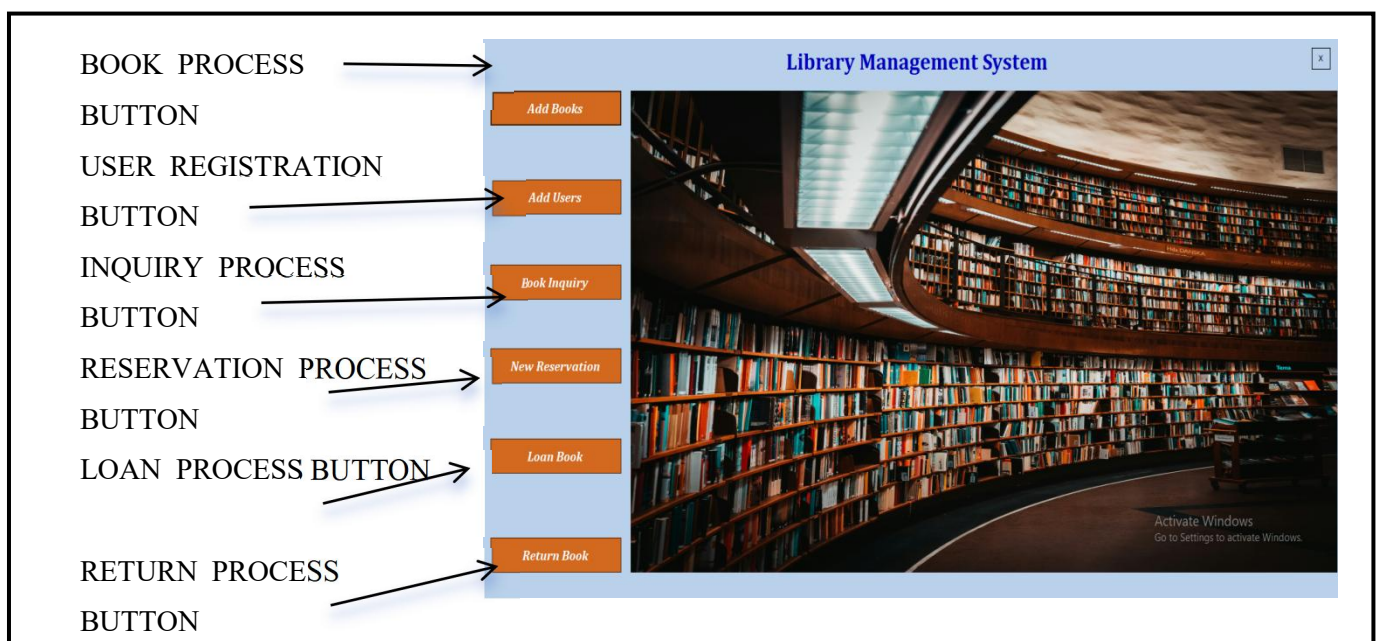
Login Form

The Login Page is the first page of the software where the admin must enter a Username and Password to access the system.

- How Does It Work:

- 1) The user opens the software and sees the login screen.
- 2) The user enters their username and password.
- 3) The user clicks the Login Button.
- 4) The system checks the entries with the SQL database table.
- 5) If correct, the system opens the Main Dashboard. If wrong, it shows an error.

✧ [Main Dashboard](#)



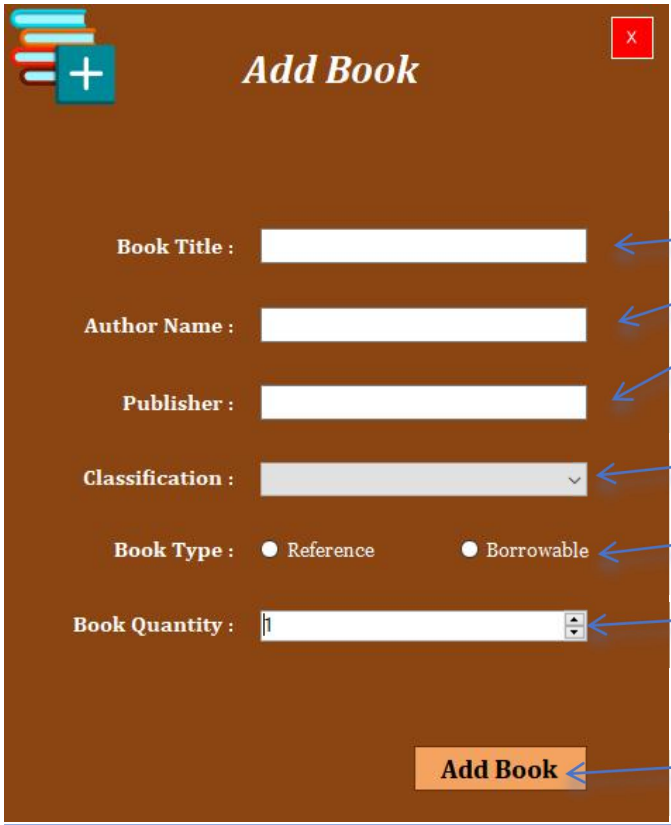
Dashboard

Serves as the central menu hub to access all sub -forms.

- How Does It Work:

Clicking any sidebar button loads the selected interface screen.

✧ Add Book Form



The image shows a web form titled "Add Book" with a brown background. It contains several input fields and controls. Annotations with arrows point to specific elements:

- TEXT BOXES:** Points to the "Book Title", "Author Name", and "Publisher" text input fields.
- COMB BOX:** Points to the "Classification" dropdown menu.
- RADIO BUTTONS:** Points to the "Reference" and "Borrowable" radio buttons under "Book Type".
- NUMERICUPDOWN BUTTON:** Points to the "Book Quantity" numeric input field.
- ADD BOOK BUTTON:** Points to the orange "Add Book" button at the bottom.

The form fields are:

- Book Title :
- Author Name :
- Publisher :
- Classification :
- Book Type : ☐ Reference ☐ Borrowable
- Book Quantity :
- Add Book

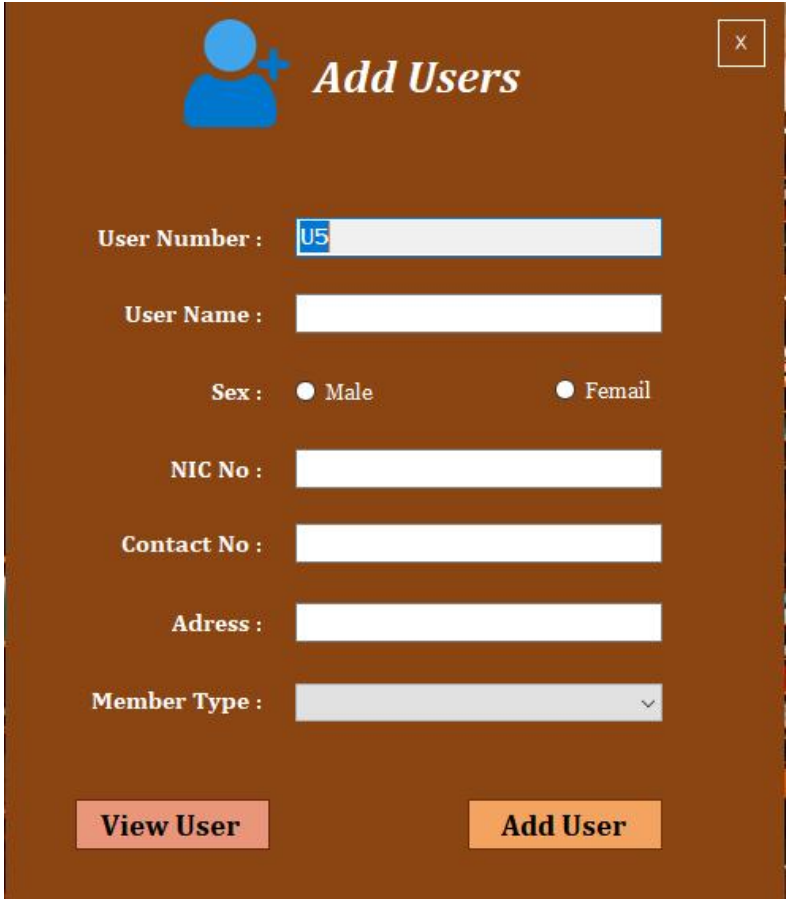
Book Registration Form

Used by the librarian to register new books into the inventory.

- How Does It Work:

The user fills the text fields and clicks "Add Book" to save data to the table.

✧ [Add Members](#)



The screenshot shows a web-based form titled "Add Users" with a blue user icon and a close button (X) in the top right corner. The form is set against a dark blue background. It contains the following fields and controls:

- User Number :** A text input field containing the value "U5".
- User Name :** An empty text input field.
- Sex :** Two radio button options: "Male" and "Female".
- NIC No :** An empty text input field.
- Contact No :** An empty text input field.
- Adress :** An empty text input field.
- Member Type :** A dropdown menu with a downward arrow icon.
- Buttons:** Two buttons at the bottom: "View User" (light blue) and "Add User" (orange).

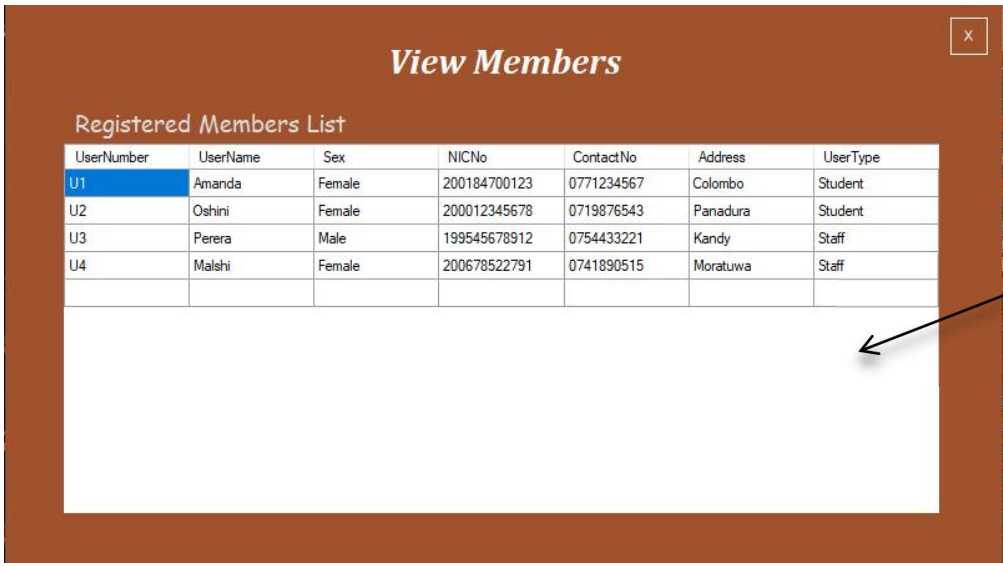
User Registration Form

Facilitates the electronic registration of new library patrons.

- **How Does It Work:**

Admin fills user details and clicks "Add User" to insert records into the Members table.

✧ [View Members](#)



View Members

Registered Members List

UserNumber	UserName	Sex	NICNo	ContactNo	Address	UserType
U1	Amanda	Female	200184700123	0771234567	Colombo	Student
U2	Oshini	Female	200012345678	0719876543	Panadura	Student
U3	Perera	Male	199545678912	0754433221	Kandy	Staff
U4	Malshi	Female	200678522791	0741890515	Moratuwa	Staff

DATAGRIDVIEW

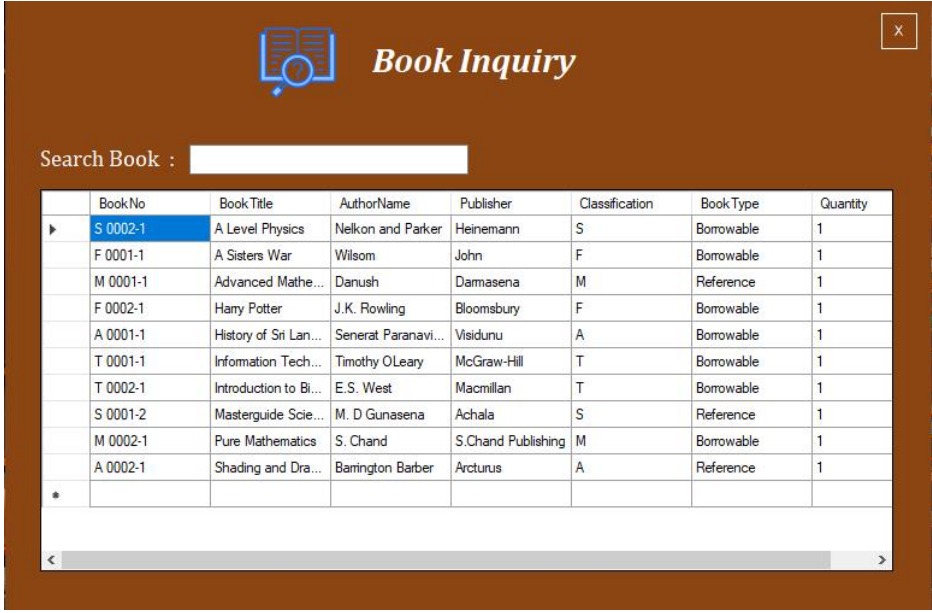
View Mebers Page

Displays an organized list of all registered library users.

● How Does It Work:

When loaded, it automatically fetches and views records from the database.

✧ [Book Inquiry Page](#)



Book Inquiry

Search Book :

	BookNo	BookTitle	AuthorName	Publisher	Classification	Book Type	Quantity
▶	S 0002-1	A Level Physics	Nelson and Parker	Heinemann	S	Borrowable	1
	F 0001-1	A Sisters War	Wilson	John	F	Borrowable	1
	M 0001-1	Advanced Mathe...	Danush	Damasena	M	Reference	1
	F 0002-1	Hary Potter	J.K. Rowling	Bloomsbury	F	Borrowable	1
	A 0001-1	History of Sri Lan...	Senerat Paranavi...	Visidunu	A	Borrowable	1
	T 0001-1	Information Tech...	Timothy OLeary	McGraw-Hill	T	Borrowable	1
	T 0002-1	Introduction to Bi...	E.S. West	Macmillan	T	Borrowable	1
	S 0001-2	Masterguide Scie...	M. D Gunasena	Achala	S	Reference	1
	M 0002-1	Pure Mathematics	S. Chand	S.Chand Publishing	M	Borrowable	1
	A 0002-1	Shading and Dra...	Barrington Barber	Arcturus	A	Reference	1
*							

Inquiry Progress

Allows the librarian to check real-time book availability.

- How Does It Work:

The grid updates instantly when keywords are typed into the search box.

✧ [Book Reservation Page](#)

The screenshot shows a web form titled "Book Reservation" with a "RESERVED" status icon. The form is divided into two main sections: "Enter Reservation Details" and "AVAILABLE USERS".

Enter Reservation Details:

- User Number :** A text input field.
- Book Number :** A text input field.
- Reservation Date :** A date picker showing "Thursday , June 4, 2026".

AVAILABLE USERS:

UserNumber	UserName
U1	Amanda
U2	Oshini
U3	

AVAILABLE BOOKS:

BookNo	BookTitle
F 0001-1	A Sisters War
F 0002-1	Harry Potter
F 0001-2	...

At the bottom of the form are two buttons: "Reserve Book" (orange) and "Cancel" (grey). Annotations with arrows point to these buttons and the date picker.

Annotations:

- DATE TIME PICKER:** Points to the "Reservation Date" field.
- RESERVATION BUTTON:** Points to the "Reserve Book" button.
- CANCLE BUTTON:** Points to the "Cancel" button.

Reservation Progress

Processes holds on library books for specific users.

- How Does It Work:

Links a user to a book and saves the booking timestamp upon clicking "Reserve Book".

✧ Loan Process Page

The screenshot shows a web application titled "Loan Process" with a "LOAN" icon. It contains input fields for "User Number" and "Book Number". Below these are date pickers for "Issue Date" (Thursday, June 4, 2026) and "Due Date" (Thursday, June 18, 2026). Two buttons, "Issue Book" (orange) and "Cancel" (grey), are positioned below the date pickers. At the bottom, there are two tables: "AVAILABLE USERS" and "AVAILABLE BOOKS".

Annotations with blue arrows point to the buttons:

- An arrow points from the text "ISSUE BOOK BUTTON" to the "Issue Book" button.
- An arrow points from the text "CANCLE BUTTON" to the "Cancel" button.

	UserNumber	UserName
▶	U1	Amanda
	U2	Oshini
	U3	

	BookNo	BookTitle
▶	F 0001-1	A Sisters War
	F 0002-1	Harry Potter
	F 0003-1	

Loan Progress

Manages the book issuing lifestyle system.

- How Does It Work:

Generates a loan record and updates the book status when the "Issue Book" button is clicked.

✧ [Return Book Page](#)



Return Book

User Number :

Book Number :

Return Date :

Fine Amount (Rs) :

CURRENTLY ISSUED BOOKS

	LoanID	UserNumber	BookNumber	IssueDate	DueDate
▶	12	U1	F 0001-1	5/20/2026	6/3/2026
	13	U1	T 0001-1	5/22/2026	6/5/2026
	14	U1	S 0002-1	5/25/2026	6/8/2026
	15	U1	A 0001-1	5/26/2026	6/9/2026
	16	U2	T 0001-1	5/1/2026	5/15/2026

Return Progress

Finalizes book returns and displays overdue fine amounts automatically.

● How Does It Work:

Compares the return date against the due date. If late, it multiplies extra days by Rs. 10 and displays the balance.

7. SUBMIT SOFTWARE DEVELOPMENT PROJECT

Project	https://drive.google.com/drive/folders/1QJNrvOktmLNRk5ohBrjPOIW-Ab63ZLW?usp=drive_link
File	